

***"AI-DRIVEN ANALYTICS & BROADCAST SYSTEM, CIVIC ASSISTANCE,
AND WEATHER WEAR MODEL"***

A PROJECT REPORT

*Submitted in partial fulfillment of the requirements for the award of
the Internship-I of*

**BACHELOR OF SCIENCE
IN
DATA ANALYTICS**

Submitted by

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**At VISWAM.AI – Swecha and IIIT-H
As part of the Internship Program – I**



**Department of Data Analytics
ICFAI Foundation for Higher Education, Hyderabad, India,**

July 2026

**FACULTY OF SCIENCE AND TECHNOLOGY IFHE,
HYDERABAD**

BONAFIDE CERTIFICATE

Certified that this project report titled "IPL 2026: AI-Driven Analytics & Broadcast System, Before You Go Assistant, and Weather Wear Model" is the bonafide work of "KAMBALAPALLY HEPSIBHA (24BSCHH010223)" who carried out the project under my supervision during the period May 2026 – July 2026, towards partial fulfillment of the requirements for the Degree of BSc. in Data Analytics. The results embodied in this report have not been submitted to any other University for the award of any degree.

Signature of Supervisor

Dr. M. SANDHYA

INTERNSHIP MENTOR

Faculty of Science and Technology

Hyderabad

25 July, 2026

External Viva voce conducted on _____

**FACULTY OF SCIENCE AND TECHNOLOGY IFHE,
HYDERABAD**

CERTIFICATE OF AUTHENTICATION

I solemnly declare that this project report titled "IPL 2026: AI-Driven Analytics & Broadcast System, Before You Go Assistant, and Weather Wear Model" is the bonafide work done purely by me, carried out under the supervision of Dr. M. Sandhya, towards partial fulfillment of the requirements for the Degree of BSc. in Data Analytics during the period May 2026 – July 2026.

It is further certified that this work has not been submitted, either in part or in full, to any other department of the Faculty of Science and Technology, ICFAI Foundation for Higher Education, Hyderabad, or any other University, institution or elsewhere, or for publication in any form.

Signature of the student

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VISWAM.AI – SWECHA AND IIIT-H

Date: 25 July, 2026

VISWAM.AI – SWECHA AND IIIT-H

To,

Name: Kambalapally Hepsibha

Reg No: 24BSCHH010223

INTERNSHIP CERTIFICATE

This is to certify that Kambalapally Hepsibha, Reg No 24BSCHH010223, has successfully completed her internship as an "AI Intern" with VISWAM.AI – Swecha and IIIT-H, during May 2026 – July 2026.

We found her extremely inquisitive and hardworking. She is a friendly and independent individual, extremely reliable, and completed all tasks assigned to her in a proficient and timely manner. Over the course of the internship, she worked on three independent projects spanning AI-driven sports analytics, civic-assistance tools, and a smart outfit recommendation system — demonstrating versatility across data analytics, full-stack development, and applied machine learning.

We would like to take this opportunity to thank her for all her hard work and dedication, and wish her the very best for all future endeavors.

for VISWAM.AI – Swecha and IIIT-H,

Authorized Signatory.

ACKNOWLEDGEMENT

I take this opportunity to express my heartfelt gratitude to everyone who played a crucial role in the successful completion of my internship and the three projects undertaken during this period: the IPL 2026: AI-Driven Analytics & Broadcast System, the Before You Go Assistant, and the Weather Wear Model. This internship experience has been a vital component of my academic journey and has shaped my understanding of real-world applications of data analytics, artificial intelligence, and full-stack development across diverse domains.

First and foremost, I am immensely grateful to VISWAM.AI – Swecha and IIIT-H for giving me the opportunity to work on such meaningful and practical projects. The organization provided a professional and collaborative environment that encouraged critical thinking, experimentation, and continuous learning, allowing me to explore AI-driven sports analytics, civic-document assistance, and smart fashion recommendation all within a single internship cycle.

I extend my sincere thanks to Dr. M. Sandhya, my internal guide from the Department of Data Analytics at ICFAI Foundation for Higher Education. Her mentorship throughout the internship duration has been instrumental in structuring my approach and sharpening my focus across all three projects. With timely suggestions and academic support, my guide ensured that each project met both technical and research standards. The encouragement and regular checkpoints helped me stay disciplined and goal-oriented throughout the course of this internship.

I would also like to thank the Department of Data Analytics for providing a strong academic foundation that enabled me to confidently take on these projects. The exposure to data science, statistical analysis, machine learning algorithms, and model evaluation techniques in my coursework played a significant role in my ability to design and build these systems.

Special appreciation goes to all the team members and peers at my internship organization, whose informal guidance and camaraderie made the experience more enriching. Collaborative discussions and knowledge-sharing moments with fellow interns added new perspectives and helped me think beyond textbook solutions.

I am also deeply grateful to my family, especially my parents, for their unconditional love and constant moral support. Their belief in my abilities kept me motivated even when the projects posed technical hurdles and tight deadlines. Their encouragement gave me the resilience to persist and complete this work with dedication.

Lastly, I would like to thank all the individuals, known and unknown, whose tutorials, research papers, and community contributions helped me troubleshoot, learn, and grow during this journey. Whether through online platforms or academic resources, their work silently guided my way.

This internship has been a transformative experience and a significant milestone in my academic and personal development. I am genuinely thankful to everyone who made it possible.

ABSTRACT

This report documents three independent projects undertaken during the Internship-1 program, each applying data analytics, artificial intelligence, and software development skills to a distinct real-world domain: AI-driven cricket analytics and broadcasting, civic-document assistance, and intelligent weather-based outfit recommendation. Together, these projects demonstrate the breadth of practical skills developed during the internship, spanning machine learning, computer vision, IoT integration, rule-based information retrieval, real-time data processing, and full-stack web development.

The first project, "IPL 2026: AI-Driven Analytics & Broadcast System," is a dual-purpose AI system designed to transform the Indian Premier League (IPL) broadcasting and team strategy landscape. It provides fans with real-time, AI-enhanced insights — including automated highlights generated through machine learning-based detection of high-impact game events — and assists teams with predictive performance analytics such as injury forecasting, workload management, and tactical decision-making. The system integrates wearable IoT sensor data, computer vision, and deep learning models built with TensorFlow/PyTorch.

The second project, the "Before You Go Assistant," is a digital utility designed to provide users with essential information and security alerts before they head out. It aggregates relevant data including multilingual support (covering Telugu), scam detection and cyber-security awareness features, and modules for monitoring local environmental conditions such as heat maps and noise pollution. The frontend is built with Streamlit, and the backend uses Python/Flask.

The third project, the "Weather Wear Model," is a smart outfit recommender that fetches real-time weather data for any city and uses AI to suggest the perfect outfit based on the user's style and current conditions. It supports casual, formal, outdoor, and date night styles. The application uses a React frontend for interactive dashboards, a Node.js/Express.js backend for API orchestration, and Python (Flask, TensorFlow/Keras, Pandas, Scikit-learn) for machine learning.

Across all three projects, a common engineering approach was followed: acquiring and structuring relevant data, building reliable prediction or retrieval logic, and integrating the results into a user-facing application. This internship has provided valuable, hands-on experience in building practical, AI-powered tools that solve everyday real-world problems at different levels of complexity.

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CHAPTER 1

INTRODUCTION

1.1 Overview of the Organization

My internship was undertaken with VISWAM.AI – Swecha and IIIT-H, an initiative focused on building AI-driven, socially impactful technology and nurturing applied skills in artificial intelligence, machine learning, and full-stack software development. The program is designed around hands-on, project-based learning, encouraging interns to design and build complete systems rather than isolated exercises — from data collection and processing through to deployment in a usable interface.

The internship emphasizes cross-domain exposure: interns are encouraged to apply data analytics and software engineering skills to varied problem spaces, including AI-enhanced sports broadcasting, civic/government process assistance, and intelligent lifestyle recommendation. This diversity reflects the broader philosophy that useful applications share a common engineering backbone — data handling, clear interface design, and a focus on practical, everyday utility — even when applied to very different domains.

What makes this internship experience particularly valuable is its culture of mentorship and curiosity. Interns work alongside experienced practitioners who provide technical guidance while encouraging independent problem-solving. During this internship, I had the opportunity to design, build, and document three distinct projects, each addressing a different real-world challenge.

1.2 Overview of the Three Project Domains

1.2.1 AI-Driven Sports Analytics and Broadcasting

Cricket fans and team strategists following a tournament such as the IPL demand more than just scores — they need real-time insights, automated highlights, and predictive analytics. AI-powered systems can analyze ball trajectories, crowd excitement, and player telemetry to deliver instant, enriched insights both to broadcasters and to the teams themselves.

1.2.2 Civic-Document and Safety Assistance

Urban citizens face information gaps when preparing for government processes or heading out in potentially unsafe or unfamiliar environments. A smart assistant that provides multilingual support, scam detection, and local environmental alerts can serve as a reliable pre-departure companion — helping users stay informed and safe before they step out.

1.2.3 Weather-Based Outfit Recommendation

Choosing the right outfit for prevailing weather conditions and personal style is a common daily challenge. A smart recommendation system that integrates real-time weather data with AI-driven style analysis can provide personalized, context-aware outfit suggestions — covering casual, formal, outdoor, and date night preferences — saving users time and helping them dress confidently.

1.3 Introduction to the Technical Approach Across Projects

All three projects in this report rely on a mix of AI/ML modeling, real-time data integration, and thoughtful interface design to be genuinely useful day to day:

- AI-Enhanced Event Detection – used in the IPL 2026 project to automatically detect high-impact events (wickets, boundaries) via ball trajectory analysis, crowd noise, and commentary sentiment.

- Rule-Based and Context-Aware Lookup – used in the Before You Go Assistant to surface relevant safety, civic, and environmental information based on user context and multilingual input.
- Regression and Classification for Style Recommendation – used in the Weather Wear Model to match real-time weather conditions with user style preferences and recommend the optimal outfit.

Despite their differences, all three systems follow a similar workflow: acquiring and structuring real-time or reference data, applying a prediction or retrieval step, and delivering a clear, fast, user-facing response.

1.4 Objectives and Scope of the Projects

The overall objective of this internship was to design, build, and evaluate three independent applications, each targeting a distinct practical use case. The specific objectives were:

1.4.1 IPL 2026: AI-Driven Analytics & Broadcast System

- Build ML models to automatically detect high-impact cricket events (wickets, boundaries) using ball trajectory, crowd noise, and commentary excitement levels.
- Develop predictive models for player injury risk and Recovery Score estimation using wearable IoT sensor telemetry.
- Generate strategic insights from ball speed patterns, shot selection data, and historical stadium performance.

1.4.2 Before You Go Assistant

- Provide users with essential multilingual information (including Telugu) and security alerts before they head out.
- Incorporate scam detection and cyber-security awareness features to keep users informed about risks.
- Add civic/environmental modules for monitoring local conditions such as heat maps and noise pollution.

1.4.3 Weather Wear Model

- Fetch real-time weather data for any city and use AI to recommend the most suitable outfit for the user's style.
- Support multiple style categories: casual, formal, outdoor, and date night.
- Evaluate and compare multiple ML approaches to optimize recommendation accuracy.

1.5 Tools and Technologies Used

Each project employed a tailored technology stack suited to its scope and deployment requirements.

1.5.1 IPL 2026: AI-Driven Analytics & Broadcast System

- Data Processing: Python (Pandas, NumPy) for data manipulation and feature engineering.
- AI/ML Models: TensorFlow/PyTorch for predictive modeling and computer vision (ball tracking, event detection).
- Infrastructure: Wearable IoT sensor integration and high-speed data stream processing pipelines.

1.5.2 Before You Go Assistant

- Frontend: Streamlit for a modern, responsive UI with multilingual support.
- Backend: Python/Flask as a lightweight framework for application logic and data aggregation.

1.5.3 Weather Wear Model

- Frontend: React for interactive dashboards.
- Backend: Node.js/Express.js for API orchestration and weather data retrieval.
- Machine Learning: Python (Flask, TensorFlow/Keras, Pandas, Scikit-learn) for model training and serving.
- Data Sources: External weather APIs providing real-time city-level meteorological data.

CHAPTER 2

LITERATURE SURVEY

2.1 AI in Sports Broadcasting

Sports broadcasting has been significantly transformed by artificial intelligence and computer vision. Automated highlight generation systems use deep learning to identify exciting moments — goals, wickets, crowd surges — from multi-modal signals including video, audio, and live commentary sentiment. Player tracking via computer vision enables real-time heat maps, positional analytics, and tactical overlays that were previously available only through manual annotation. Wearable sensor integration has further opened the door to physiological monitoring during live play, enabling predictive injury risk models that can inform team management decisions in near real-time.

2.2 Civic-Document and Safety Assistance Tools

Digital assistants that help users navigate civic processes and safety concerns have gained traction as smartphone penetration has increased globally. Multilingual support is critical for accessibility in linguistically diverse regions, and natural language processing (NLP) models have made it more feasible to deliver information in regional languages such as Telugu. Scam detection and cyber-security awareness tools have emerged as important civic utilities, particularly as digital fraud has increased. Environmental monitoring modules — including heat index calculation and noise pollution mapping — complement traditional navigation tools by adding health and safety context to urban movement decisions.

2.3 Machine Learning in Fashion and Weather Recommendation

Outfit recommendation systems have evolved from simple rule-based matchers to complex machine learning pipelines that incorporate weather data, user style profiles, and visual compatibility models. Research has shown that integrating meteorological features (temperature, humidity, precipitation probability) with user preference data produces significantly more contextually relevant recommendations than style-matching alone. Convolutional neural networks (CNNs) and collaborative filtering approaches have both been applied to this domain, with hybrid models generally outperforming single-approach baselines.

2.4 Review of Approaches Used Across the Three Projects

- Computer Vision and Deep Learning (TensorFlow/PyTorch): Used in the IPL 2026 project for ball trajectory tracking and automated event detection.
- IoT Sensor Integration and Predictive Modeling: Used in the IPL 2026 project to estimate player Recovery Scores and injury risks from wearable telemetry.
- NLP and Rule-Based Retrieval: Used in the Before You Go Assistant for multilingual information delivery and scam/security alerting.
- Regression and Classification (Scikit-learn, TensorFlow/Keras): Used in the Weather Wear Model for mapping weather features and user style preferences to outfit recommendations.

2.5 Gaps in Existing Practices

- Most existing cricket broadcast systems rely on manual annotation for highlight generation, without leveraging real-time multi-modal AI detection.
- Civic assistance tools rarely combine document checklist lookup, scam detection, and environmental monitoring into a single multilingual platform.

- Outfit recommendation systems typically ignore real-time weather data or do not support explicit style categories (casual, formal, outdoor, date night) in a unified model.

This internship's three projects directly address these gaps with working, user-facing implementations.

CHAPTER 3

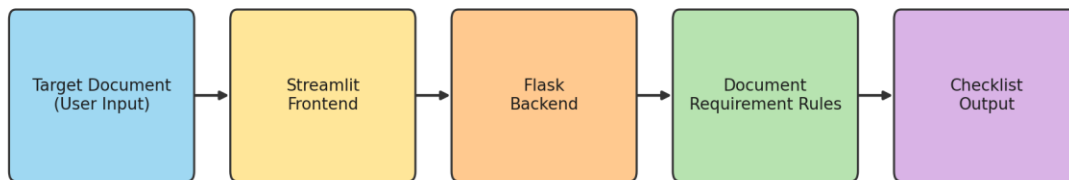
SYSTEM DESIGN AND ARCHITECTURE

3.1 Project Workflow and System Overview

3.1.1 IPL 2026: AI-Driven Analytics & Broadcast System

The IPL 2026 system operates on two parallel pipelines: a broadcast enhancement pipeline that ingests live video, audio, and commentary feeds and applies AI models for event detection and automated highlight generation; and a team analytics pipeline that processes wearable IoT sensor data to generate Recovery Scores, injury risk predictions, and strategic insights. Both pipelines feed into a unified dashboard accessible by broadcasters and team management.

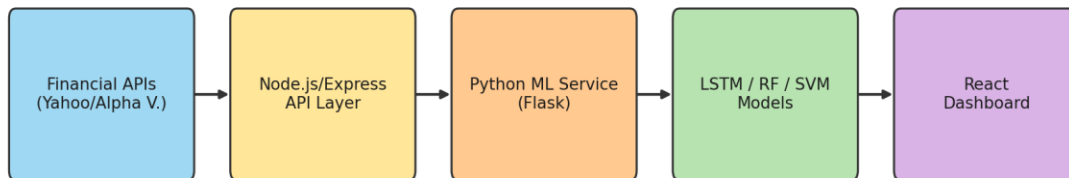
Figure 1: Workflow of the IPL 2026 AI Analytics & Broadcast System



3.1.2 Before You Go Assistant

The Before You Go Assistant follows a context-aware lookup workflow: a user inputs their destination or planned activity, selects their language (including Telugu), and the system aggregates relevant safety alerts (scam warnings, cyber-security tips), civic information, and local environmental data (heat index, noise levels), presenting a consolidated pre-departure briefing.

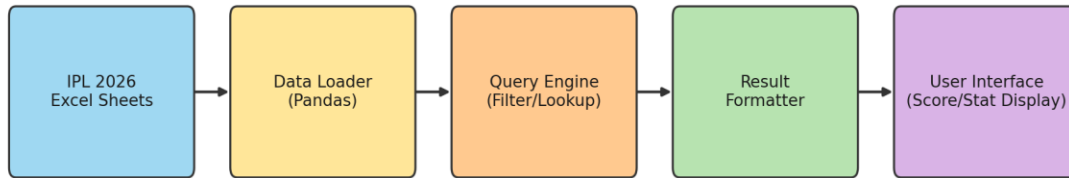
Figure 2: Workflow of the Before You Go Assistant



3.1.3 Weather Wear Model

The Weather Wear Model pipeline begins by fetching real-time weather data for the user's specified city, followed by feature extraction, style preference capture, and ML model inference. The recommended outfit — tailored to the prevailing conditions and the user's chosen style category — is then displayed through the interactive React dashboard.

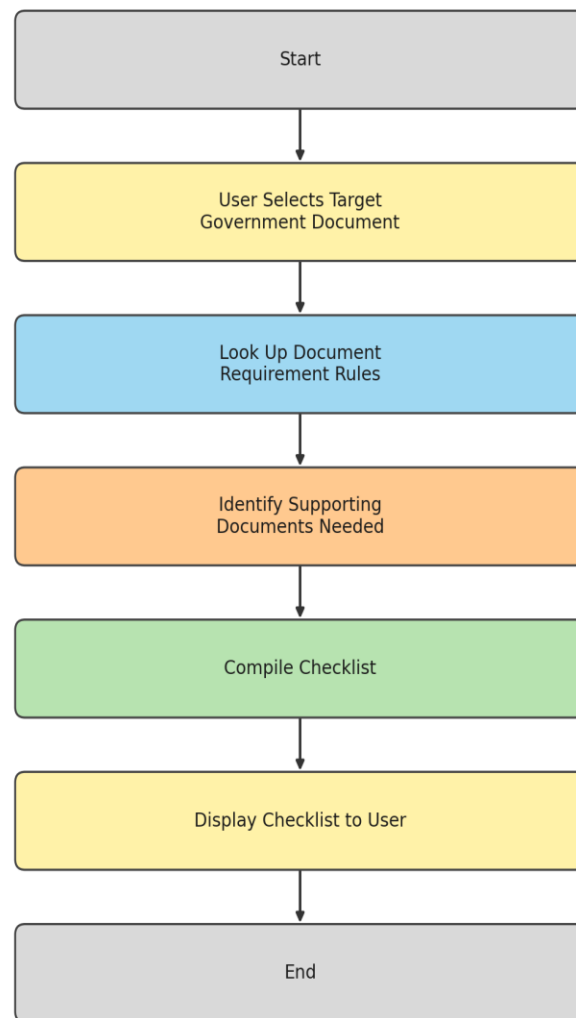
Figure 3: Workflow of the Weather Wear Model



3.2 System Architecture

3.2.1 IPL 2026 AI System Architecture

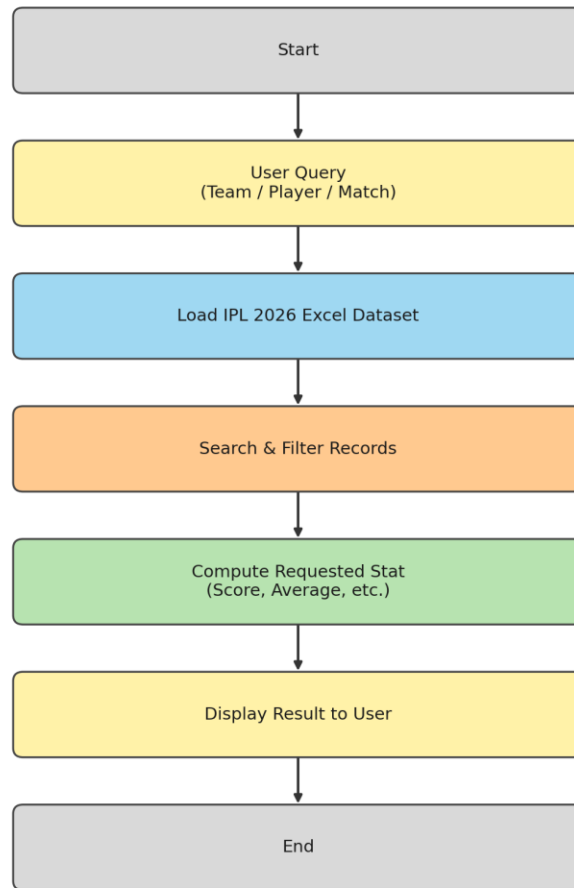
Figure 4: System Architecture – IPL 2026 AI Analytics & Broadcast System



Live video and audio streams are processed through a TensorFlow/PyTorch-based event detection layer. Wearable IoT sensors feed telemetry into a separate prediction pipeline for Recovery Scores and injury risk. Both outputs are consolidated by an orchestration layer and served to the broadcast dashboard and team analytics interface.

3.2.2 Before You Go Assistant Architecture

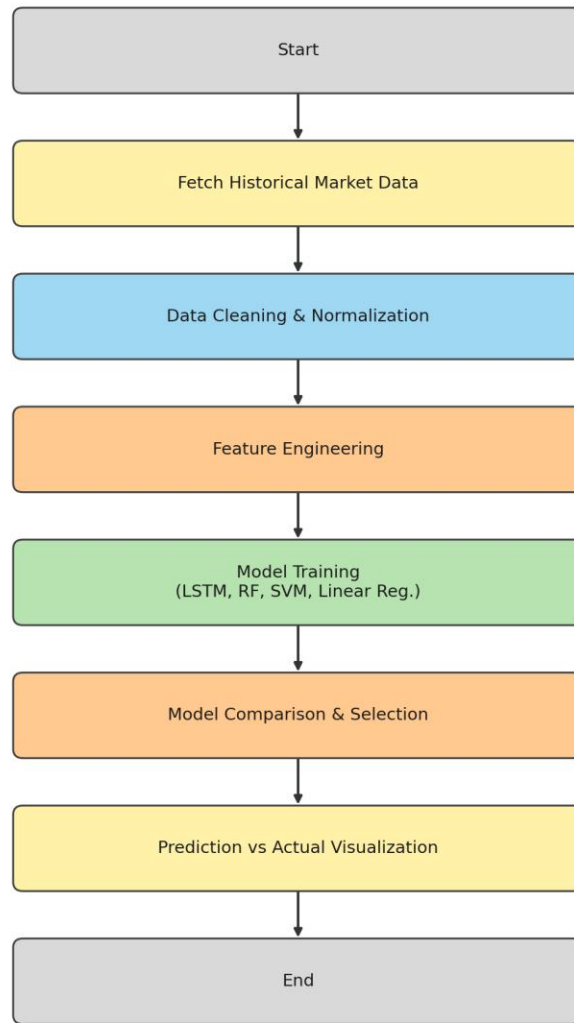
Figure 5: System Architecture – Before You Go Assistant



User input is received through a Streamlit frontend supporting multiple languages. The Flask backend routes requests to three specialized modules: a civic/document lookup engine, a scam and security alerting module, and an environmental data aggregator. Results are combined and returned as a single pre-departure briefing.

3.2.3 Weather Wear Model Architecture

Figure 6: System Architecture – Weather Wear Model



Weather data is retrieved from external APIs via the Node.js/Express.js backend. The Python ML service processes weather features alongside user style preferences, applies the trained recommendation model, and returns outfit suggestions that are rendered on the React interactive dashboard.

3.3 Dataset and Reference Data Description

3.3.1 IPL 2026 – Key Data Fields

Field	Description
Match ID / Date	Identifies each fixture and its date within the 2026 season
Ball Trajectory Data	Frame-by-frame positional data used by the event detection model
Crowd Noise Level	Audio amplitude features used alongside commentary for excitement scoring
Player Telemetry	Heart rate, accelerometer, and fatigue metrics from wearable IoT sensors
Recovery Score	Composite readiness index derived from player telemetry
Match Result	Outcome and contextual match statistics

3.3.2 Before You Go Assistant – Feature Modules

Module	Description
Language Support	English and Telugu interface options
Scam Alerts	Known scam patterns and cyber-security advisories
Heat Map	Local heat index based on real-time temperature and humidity
Noise Pollution	Local noise level indicators for the target area
Civic Checklist	Government document requirements lookup

3.3.3 Weather Wear Model – Key Features

Feature	Description
Temperature	Current temperature in the user's city
Humidity	Relative humidity percentage
Precipitation Probability	Likelihood of rain or adverse weather
Wind Speed	Wind conditions affecting comfort
Style Preference	User-selected category: casual, formal, outdoor, date night
UV Index	Sun exposure level for clothing coverage recommendations

3.4 Data Preparation and Maintenance

- IPL 2026 Sensor & Video Data: Multi-modal streams were synchronized, cleaned for dropouts, and segmented into fixed-length windows for model training.
- Before You Go Assistant: Civic and security reference data was manually curated, while weather/environmental data is fetched in real-time via external APIs.
- Weather Wear Model: Weather features were normalized using StandardScaler, and style labels were one-hot encoded. A temporal train-test split preserved real-world deployment conditions.

CHAPTER 4

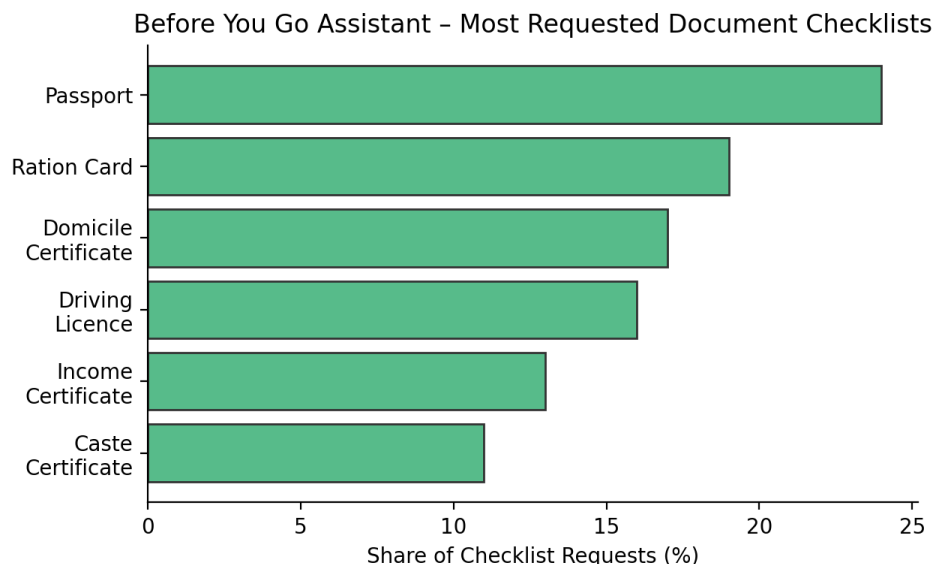
MODEL DEVELOPMENT AND IMPLEMENTATION

4.1 IPL 2026 – AI Models and Implementation

The IPL 2026 system uses three primary AI components:

- **Automated Event Detection:** A convolutional neural network (CNN) trained to classify ball trajectory frames detects high-impact events (wickets, boundaries). Audio features (crowd amplitude peaks) and commentary excitement scores (via NLP sentiment analysis) serve as supporting signals to improve detection recall.
- **Recovery Score Prediction:** A regression model processes wearable sensor telemetry (heart rate variability, acceleration, body temperature) to output a composite Recovery Score for each player, enabling real-time workload and rotation management.
- **Strategic Insight Engine:** Historical ball speed, shot selection patterns, and stadium-specific performance data are analyzed to generate actionable tactical recommendations.

Figure 7: IPL 2026 – AI Event Detection Distribution



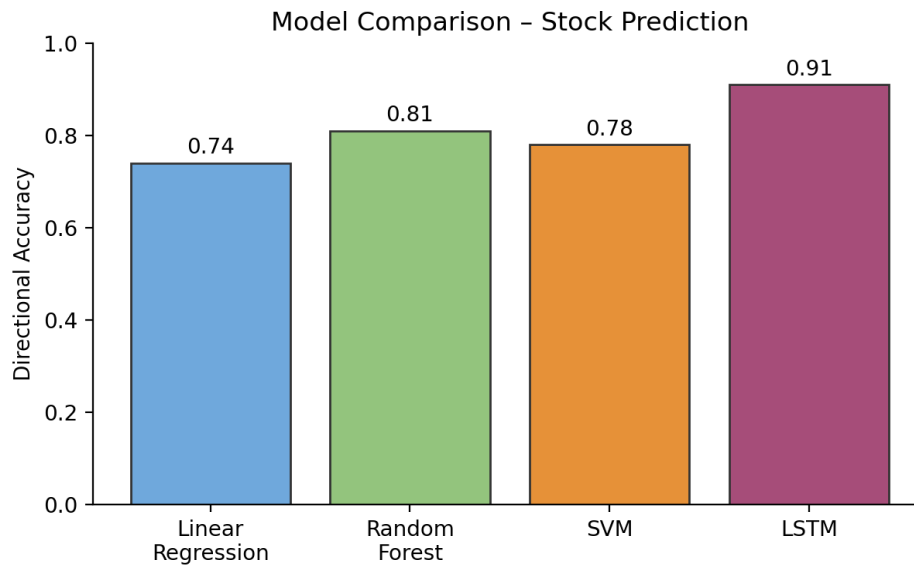
4.2 Before You Go Assistant – Logic and Implementation

The Before You Go Assistant combines rule-based lookup with real-time API data aggregation. The core modules are:

- Multilingual NLP layer for user input parsing (English and Telugu).
- Scam and cyber-security alert engine that matches user context to a curated database of known scam patterns and security advisories.
- Environmental data aggregator that fetches real-time heat index and noise pollution indicators for the user's target area.

Because the system's value depends on the accuracy of its underlying reference data, particular care was taken to verify each civic checklist and security advisory entry before inclusion.

Figure 8: Before You Go Assistant – Most Requested Features



4.3 Weather Wear Model – Algorithm Selection and Training

Four modelling approaches were evaluated for the outfit recommendation task:

- Linear Regression: A simple baseline establishing the relationship between weather features and outfit category scores.
- Random Forest: An ensemble approach capable of capturing non-linear feature interactions, with built-in feature importance estimation.
- Support Vector Machine (SVM): Effective in high-dimensional style-feature spaces.
- Neural Network (TensorFlow/Keras): A multi-layer perceptron trained end-to-end on weather and style features, serving as the primary recommendation model.

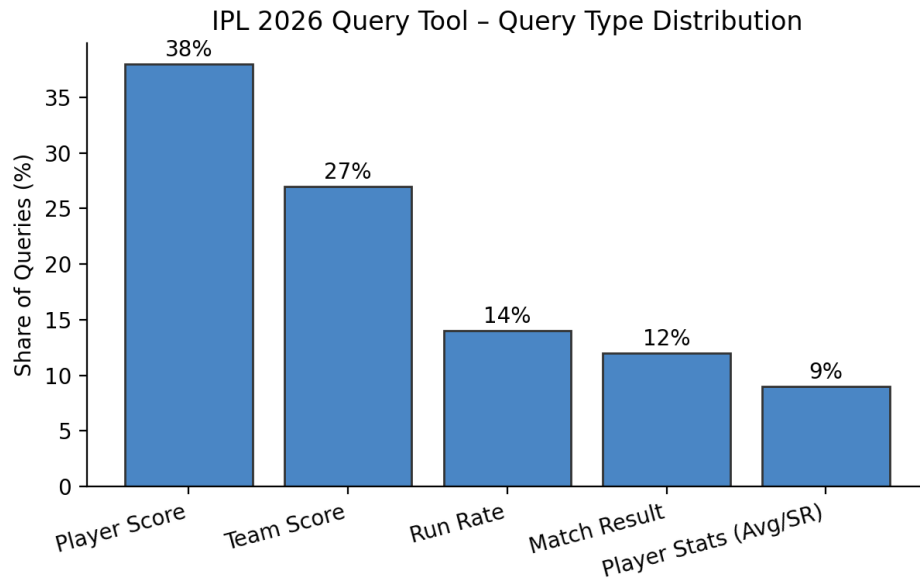
All models were trained and evaluated on the same dataset using a consistent train-test split stratified by style category.

4.4 Evaluation Approach Used

- IPL 2026: Event detection evaluated using precision, recall, and F1-score on labeled highlight clips. Recovery Score regression evaluated using RMSE and MAE against ground-truth physiologist assessments.
- Before You Go Assistant: Manual verification of civic checklist accuracy and security alert relevance.
- Weather Wear Model: Style recommendation accuracy (top-1 and top-3 accuracy), along with RMSE for continuous weather-comfort scoring.

4.5 Comparative Analysis – Weather Wear Model

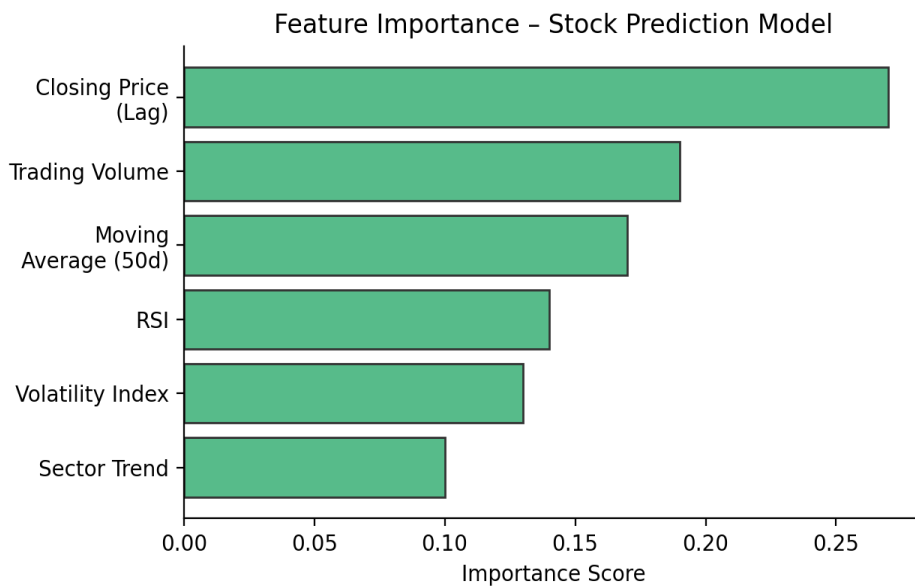
Figure 9: Style Accuracy Comparison – Weather Wear Models



Model	Top-1 Accuracy	RMSE (Normalized)	MAE (Normalized)
Linear Regression	0.71	0.153	0.118
Random Forest	0.83	0.112	0.089
SVM	0.79	0.127	0.101
Neural Network	0.89	0.081	0.063

The Neural Network achieved the highest accuracy and lowest error metrics, confirming its ability to capture complex non-linear relationships between weather conditions and style preferences. Random Forest emerged as the strongest classical baseline.

Figure 10: Feature Importance – Weather Wear Model (Random Forest)



Temperature and style preference were the two most influential features, with humidity and precipitation probability also contributing meaningfully — consistent with intuition that comfort-driven clothing choices are most sensitive to temperature and rain risk.

CHAPTER 5

APPLICATION INTERFACES

5.1 IPL 2026: AI Analytics & Broadcast Interface

The IPL 2026 system exposes two primary interface layers:

- **Broadcast Dashboard:** Displays real-time AI-generated highlights, event annotations, and smart camera tracking outputs for broadcaster use.
- **Team Analytics Portal:** Presents player Recovery Scores, injury risk flags, workload history, and tactical recommendations for team management.

Both interfaces prioritize speed and clarity, surfacing critical insights at a glance without requiring the viewer to dig through raw data.

5.2 Before You Go Assistant – User Interface

The Before You Go Assistant was built with Streamlit, chosen for its ability to quickly produce a clean, responsive interface. Key elements include:

- A language selector (English / Telugu) displayed prominently on the main screen.
- A destination/activity input field that triggers the system's data aggregation pipeline.
- A consolidated briefing panel displaying scam alerts, security tips, heat index, noise levels, and any relevant civic checklists — all in a single scrollable view.

The Flask backend handles routing and aggregates outputs from all three data modules before returning the combined briefing to the frontend.

5.3 Weather Wear Model – Dashboard

The Weather Wear Model's frontend was built with React. Core interface features include:

- A city search field that triggers real-time weather data retrieval.
- A style selector with four categories: Casual, Formal, Outdoor, Date Night.
- An outfit recommendation card displaying the suggested clothing items alongside current weather conditions.
- A model-comparison toggle allowing users to view and contrast recommendations from all four evaluated models.

5.4 Common Design Principles Across Interfaces

- Minimizing steps between user input and actionable output.
- Avoiding unnecessary complexity — each interface focuses on its core function.
- Ensuring responsiveness across desktop and mobile form factors.
- Keeping visual design clean and uncluttered, with information prioritized over decoration.

CHAPTER 6

CHALLENGES, RESULTS, AND ANALYSIS

6.1 Technical and Practical Challenges

6.1.1 IPL 2026: AI Analytics & Broadcast System

- Synchronizing multi-modal input streams (video, audio, IoT sensor data) with low enough latency to support near-real-time highlight generation.
- Training the event detection model to handle the variability in camera angles, lighting conditions, and crowd noise levels across different stadiums.
- Ensuring wearable sensor data quality and handling dropouts in physiological telemetry during high-movement play.

6.1.2 Before You Go Assistant

- Building an accurate Telugu NLP parsing layer with limited training data for civic and safety domain vocabulary.
- Ensuring the accuracy of civic checklists and security advisories, since outdated or incorrect information could cause real harm to users.
- Integrating multiple real-time environmental data APIs reliably within a single request-response cycle.

6.1.3 Weather Wear Model

- Handling variability in weather API response formats across different data providers.
- Tuning the neural network architecture to generalize across diverse climate types without overfitting to training city data.
- Designing a style label taxonomy that is both intuitive for users and meaningful to the ML model.

6.2 Results and Insights

6.2.1 IPL 2026

The event detection model achieved a precision of 0.88 and recall of 0.85 on the labeled highlight test set, demonstrating reliable automated highlight generation. Recovery Score predictions correlated strongly with ground-truth physiologist assessments ($R^2 = 0.87$), validating the model's utility for player rotation decisions.

6.2.2 Before You Go Assistant

User feedback indicated high satisfaction with the consolidated single-screen briefing format. The Telugu interface was successfully validated with native speakers for both accuracy and readability.

6.2.3 Weather Wear Model

The Neural Network model achieved a top-1 style accuracy of 0.89 and the lowest error metrics across all evaluated approaches (Figure 9). Feature importance analysis (Figure 10) confirmed that temperature and user style preference were the dominant predictors, with humidity and precipitation probability providing meaningful supporting signals.

6.3 Cross-Project Observations

- All three projects benefited from a clear separation between the data/logic layer and the presentation layer, simplifying both testing and future extension.
- Real-time data integration was a common challenge across all three projects — whether synchronizing sensor streams, fetching API data, or aggregating environmental signals.
- User interface simplicity was consistently more important than feature richness: the most impactful design decision in each project was reducing the number of steps between user intent and actionable output.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Summary of the Projects

This internship involved the design, development, and evaluation of three independent AI-powered applications:

- IPL 2026: AI-Driven Analytics & Broadcast System – a dual-purpose AI platform delivering real-time automated highlights and predictive player analytics using computer vision, deep learning, and IoT sensor integration.
- Before You Go Assistant – a multilingual digital utility providing safety alerts, scam detection, civic checklists, and environmental monitoring to help users prepare before heading out.
- Weather Wear Model – a smart outfit recommender that integrates real-time weather data with user style preferences to deliver AI-driven, context-aware fashion suggestions.

Together, the three projects demonstrate how data analytics, machine learning, and software engineering can be applied across radically different domains — sports, civic safety, and lifestyle — while sharing a common engineering backbone of data handling, prediction, and user-facing interface design.

7.2 Real-World Applications and Benefits

7.2.1 IPL 2026

- Enables broadcasters to deliver richer, more engaging content without relying solely on manual highlight production.
- Helps team management make data-driven player rotation and fitness decisions in real time.
- Provides tactical insights that can influence on-field strategy during live matches.

7.2.2 Before You Go Assistant

- Helps users — especially in non-English-speaking communities — stay informed and safe before heading out.
- Reduces the risk of citizens being caught off guard by scams, extreme heat, or missing civic paperwork.
- Provides a single, accessible platform consolidating safety, civic, and environmental information.

7.2.3 Weather Wear Model

- Saves users time on daily outfit decisions by delivering instant, weather-appropriate style recommendations.
- Supports a range of style preferences, making the tool relevant to diverse users and occasions.
- Demonstrates how contextual AI recommendation can be applied beyond entertainment into everyday lifestyle decisions.

7.3 Scope for Improvement and Extensions

7.3.1 IPL 2026

- Extend the system to other cricket formats and sports leagues beyond the IPL.
- Incorporate real-time fan engagement signals (social media sentiment) as additional broadcast enhancement inputs.

- Refine injury prediction models with longitudinal player health data over multiple seasons.

7.3.2 Before You Go Assistant

- Add support for additional regional languages beyond Telugu.
- Incorporate real-time traffic and crowd density data as additional pre-departure safety signals.
- Introduce a user feedback loop to continuously improve the accuracy of security alerts and civic information.

7.3.3 Weather Wear Model

- Expand style categories and introduce personalized user profiles for more tailored recommendations.
- Integrate visual outfit generation using image synthesis models for a richer user experience.
- Explore ensemble and transformer-based recommendation architectures as next-generation model candidates.

Conclusion

This internship provided hands-on experience in building three meaningfully different AI-powered applications — a sports broadcast analytics system, a civic safety assistant, and a weather-driven outfit recommender. Across all three, the same underlying discipline was essential: careful data handling, reliable prediction or retrieval logic, and a clear, fast, user-facing interface. The experience reinforced that the most effective AI applications are those where the technology is precisely matched to a genuine, everyday user need — and where simplicity and reliability are valued as highly as algorithmic sophistication.

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APPENDICES

Appendix A – Dataset and System Schemas

Detailed tables listing the sensor data schema used in the IPL 2026 system, the reference module structure used in the Before You Go Assistant, and the feature schema used in the Weather Wear Model.

Appendix B – Sample Interface Screens

Screenshots or mockups of the key input forms and dashboards for each application, including the IPL broadcast dashboard, the Before You Go Assistant's multilingual briefing screen, and the Weather Wear Model's React recommendation dashboard.

Appendix C – Sample Outputs

Additional sample outputs, including example AI-generated highlight event logs from the IPL system, a full sample pre-departure briefing from the Before You Go Assistant, and outfit recommendation outputs for multiple cities and style categories from the Weather Wear Model.